

In [17]:

```
'''
1.Create a list and perform the following methods
a) insert() b) remove() c) append() d) len() e) pop() f) clear()
'''
```

```
myList = ["I", "like", "this", "day",]
print('my list is',myList)
myList.insert(0,"Yes")
print('my list is after insertion',myList)
myList.remove("Yes")
print('my list is after deletion',myList)
myList.append(["a", "true"])
print('my list is after appending',myList)
print('length of my list',len(myList))
myList.pop()
'sun'
print('my list is after popping',myList)
myList.clear()
print('List after clear():',myList)
```

```
my list is ['I', 'like', 'this', 'day']
my list is after insertion ['Yes', 'I', 'like', 'this', 'day']
my list is after deletion ['I', 'like', 'this', 'day']
my list is after appending ['I', 'like', 'this', 'day', ['a', 'true']]
length of my list 5
my list is after popping ['I', 'like', 'this', 'day']
List after clear(): []
```

In [18]:

```
'''
2.Develop a python program to implement Linear Search
'''
```

```
def linear_Search(list1, n, key):
    for i in range(0, n):
        if (list1[i] == key):
            return i
    return -1
list1 = [1 ,3, 5, 4, 7, 9]
key = 1
n = len(list1)
res = linear_Search(list1, n, key)
if(res == -1):
    print("Element not found")
else:
    print("Element found at index: ", res)
```

```
Element found at index: 0
```

In [19]:

```
'''  
3.Develop a python program to implement Bubble Sort  
'''  
  
def bubble_sort(list1):  
    for i in range(0,len(list1)-1):  
        for j in range(len(list1)-1):  
            if(list1[j]>list1[j+1]):  
                temp = list1[j]  
                list1[j] = list1[j+1]  
                list1[j+1] = temp  
    return list1  
list = [5, 3, 8,4,5, 7, 2]  
print("The unsorted list is: ", list)  
print("The sorted list is: ", bubble_sort(list))
```

The unsorted list is: [5, 3, 8, 4, 5, 7, 2]

The sorted list is: [2, 3, 4, 5, 5, 7, 8]

In [20]:

```
'''
4.Using Regular Expressions, develop a Python program to
a) Identify a word with a sequence of one upper case letter followed by Lower
case Letters.
b) Find all the patterns of "1(0+)1" in a given string.
c) Match a word containing 'z' followed by one or more o's.
Prompt the user for input
'''
```

```
#A
import re
def text_match(text):
    patterns = '[A-Z]+[a-z]+$'
    if re.search(patterns, text):
        return 'Found a match!'
    else:
        return('Not matched!')
print(text_match("AaBbGg"))
print(text_match("python"))

#B
def pattern(str):
    last = str[0]
    i = 1; counter = 0
    while (i < len(str)):
        if (str[i] == '0' and last == '1'):
            while (str[i] == '0'):
                i += 1
            if (str[i] == '1'):
                counter += 1
        last = str[i]
        i +=1
    return counter
str = "1001ab010abc01001"
ans = pattern(str)
print(ans)

#C
import re
def text_match(text):
    patterns = '\w*z.\w*'
    if re.search(patterns, text):
        return 'Found a match!'
    else:
        return('Not matched!')

print(text_match("ohoo"))
print(text_match("hii"))
```

```
Found a match!
Not matched!
2
Not matched!
Not matched!
```

In [21]:

```
'''
5.Develop a Python program to implement the Rock-Paper-Scissor game.
'''
import random

user_action = input("Enter a choice (rock, paper, scissors): ")
possible_actions = ["rock", "paper", "scissors"]
computer_action = random.choice(possible_actions)
print(f"\nYou chose {user_action}, computer chose {computer_action}.\n")

if user_action == computer_action:
    print(f"Both players selected {user_action}. It's a tie!")
elif user_action == "rock":
    if computer_action == "scissors":
        print("Rock smashes scissors! You win!")
    else:
        print("Paper covers rock! You lose.")
elif user_action == "paper":
    if computer_action == "rock":
        print("Paper covers rock! You win!")
    else:
        print("Scissors cuts paper! You lose.")
elif user_action == "scissors":
    if computer_action == "paper":
        print("Scissors cuts paper! You win!")
    else:
        print("Rock smashes scissors! You lose.")
```

You chose paper, computer chose paper.

Both players selected paper. It's a tie!

In [22]:

```
'''
6. Write a python program to check whether the given string is palindrome or not.
'''

st = input("enter string : ")

if(st == st[::-1]):
    print("This is a Palindrome String")
else:
    print("This is Not")
```

This is Not

In [23]:

```
'''
7. Write a python program to find factorial of a given number using functions
'''
def factorial(num):
    fact=1
    for i in range(1, num+1):
        fact=fact*i
    return fact
number=int(input("Please enter any number to find factorial: "))
result=factorial(number)
print("The factorial of %d = %d"%(number,result))
```

The factorial of 10 = 3628800

In [24]:

```
'''
8. Write a Python function that takes two lists and returns True if they are equal otherwise
false
'''

# Python 3 code to demonstrate
# check if list are identical
# using sort() + == operator

# initializing lists
test_list1 = [1, 2, 4, 3, 5]
test_list2 = [1, 2, 4, 3, 5]

# printing lists
#print("The first list is : " +str(test_list1))
#print("The second list is : " +str(test_list2))

# sorting both the lists
test_list1.sort()
test_list2.sort()

# using == to check if
# lists are equal
if test_list1 == test_list2:
    print("The lists are identical")
else:
    print("The lists are not identical")
```

The lists are identical